# Beyond Associativity and Alternativity in 16D Sedenion Signal Processing: Mutually-Annihilating Filter Cascades, the Cubic Alternativity-Defect Calculus $N_u$, and Closed-Form Pontryagin Optimal Control

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***Abstract*—In every associative algebra—including the real field $\mathbb{R}$, the complex field $\mathbb{C}$, the quaternion division ring $\mathbb{H}$, and the matrix ring $M_n(\mathbb{C})$—two linear filters $w_1, w_2$ that annihilate each other ($w_1 \cdot w_2 = 0$) produce identically zero output when cascaded on any input signal $x$, since $(x \cdot w_1) \cdot w_2 = x \cdot (w_1 \cdot w_2) = 0$. Similarly, in classical array processing, inverting or equalizing a 16-dimensional channel operator $L(u)$ requires $\mathcal{O}(16^3)$ Gaussian elimination or Singular Value Decomposition (SVD). In this paper, we show that crossing the Hurwitz and Moufang thresholds into the 16-dimensional Cayley–Dickson sedenion algebra $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$ overturns both paradigms through five exact algebraic laws, all machine-checked with *zero axioms* in the Lean 4 theorem prover (`204/204` theorems): (i) *The Symmetric Alternativity-Defect Calculus:* left and right alternativity fail by the exact same symmetric linear operator $N_u(y) \triangleq u(uy) - u^2 y = (yu)u - yu^2 = ([a,b,d]_{\mathbb{O}}, -[a,b,c]_{\mathbb{O}})$, which governs the normal operator $L(u)^\top L(u) = \|u\|_2^2 I_{16} - N_u$ and satisfies the universal cubic minimal polynomial $N_u^3 = 4\Lambda(u)^2 N_u$. Consequently, exact channel inversion $L(u)^{-1}$, Tikhonov/MMSE equalization, Moore–Penrose pseudoinversion $L(u)^+$ on the singular zero-divisor cone, and orthogonal projection $P_{\ker L(u)}$ collapse to 3-term polynomials in $N_u$ requiring only two octonion associators and zero SVD; (ii) *The Co-Axial Multi-Jammer Theorem and Mutually-Annihilating Cascades:* the 84 primitive zero-divisor blades partition into 14 box-kite sextets over $S^6$ by their native associator axis $n(u) = (a' \times_{\mathbb{O}} b')/\Lambda(u)$, and $\dim(\ker L(u_1) \cap \ker L(u_2)) \in \{0, 2, 4\}$ is non-zero if and only if $n(u_1) = +n(u_2)$. When two jammers are non-coaxial ($\ker L(u_1) \cap \ker L(u_2) = \{0\}$), we construct a 2-stage cascade $(w_1, w_2)$ satisfying $w_1 \cdot w_2 = 0$ that annihilates every linear combination $J = \alpha u_1 + \beta u_2$ while transmitting desired signals strictly via the non-associative associator $(s \cdot w_1) \cdot w_2 = [s, w_1, w_2] \neq 0$; (iii) *Pontryagin's Maximum Principle over* $\mathbb{S}$*:* maximizing the control Hamiltonian $\mathcal{H}(x, p, J, w) = \langle p, x \cdot w \rangle - \frac{\gamma}{2}\|J \cdot w\|_2^2 - \frac{\rho}{2}\|w\|_2^2$ yields a strictly concave quadratic program whose global maximizer $w^*$ is given in closed 3-term SVD-free form by $N_J$ applied to the Cayley–Dickson switching vector $\psi = x^* \cdot p$; (iv) *Eisenstein* $S_3$ *Phase Automorphisms and the Sextic Invariant* $J_3(u)$*:* we prove that the $S_3$ factor in $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$ arises from cube-root-of-unity phase rotations $Z \mapsto e^{2\pi i/3} Z$ ($\bar{\omega}^2 = \omega \iff \omega^3 = 1$) and construct a sextic $G_2 \times S_3$ invariant $J_3(u) = \mathrm{Re}(q^3)/|q|^3$ that separates lower-octonion ($J_3 = +1$) from twisted-sector ($J_3 = -1$) excitations sharing identical singular spectra; and (v) *Hurwitz–Radon 9-Frame Isometries and Extremal* $[16, 8, 6]_3$ *Ternary FEC:* the 9D subspace $\mathrm{span}\{e_0, \ldots, e_8\}$ preserves Euclidean norm against arbitrary zero-divisor keys and reduces modulo 3 to an extremal self-dual $[16, 8, 6]_3$ ternary code correcting all 1- and 2-trit errors on a 64 MHz Cortex-M4F microcontroller.**

***Index Terms*—Sedenion algebra, alternativity defect, non-associative filter cascades, Pontryagin's Maximum Principle, box-kite zero divisors, $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$, Hurwitz–Radon 9-frames, extremal $[16, 8, 6]_3$ self-dual ternary codes, Lean 4 formal verification.**

## I. Introduction: Five Paradoxes Beyond the Hurwitz and Moufang Thresholds

HURWITZ'S celebrated 1898 composition theorem establishes that $\mathbb{R}$ ($\dim 1$), $\mathbb{C}$ ($\dim 2$), $\mathbb{H}$ ($\dim 4$), and $\mathbb{O}$ ($\dim 8$) are the only normed division algebras over the real numbers [1]. At each step of the Cayley–Dickson doubling tower $\mathbb{A}_n = \mathbb{A}_{n-1} \oplus \mathbb{A}_{n-1} e_{2^{n-1}}$, a classical algebraic symmetry is surrendered: $\mathbb{C} \to \mathbb{H}$ surrenders commutativity ($xy \neq yx$); $\mathbb{H} \to \mathbb{O}$ surrenders associativity ($[x, y, z] \triangleq (xy)z - x(yz) \neq 0$) while retaining *alternativity* ($[x, x, y] = [y, x, x] = 0$) and the Moufang identities; and $\mathbb{O} \to \mathbb{S} = \mathbb{A}_4$ ($\dim 16$) surrenders alternativity ($u(uy) \neq u^2 y$) and norm composition ($\|uy\|_2 \neq \|u\|_2 \|y\|_2$), giving birth to Moreno's 11-dimensional Stiefel cone of unit-norm zero divisors [2].

In the mathematical physics and signal processing literature, the loss of alternativity at $\dim 16$ has long been viewed as an algebraic catastrophe: because $u^*(uy) \neq \|u\|_2^2 y$, standard Cayley–Dickson division $u^{-1} = u^*/\|u\|_2^2$ fails to invert the left multiplication operator $L(u) : y \mapsto u \cdot y$. Yet when we examine the operator structure of $\mathbb{S}$ from first principles, what appears to be algebraic pathology turns out to be a remarkably rigid, finite-degree operator calculus with capabilities that are *provably impossible* in any associative or alternative algebra.

Specifically, this paper resolves five foundational questions at the intersection of non-associative algebra, optimal control, and embedded radio-frequency (RF) signal processing:

1) **The Mutually-Annihilating Cascade Paradox:** Can two linear filter stages $w_1, w_2$ that annihilate each other

Manuscript completed November 2026. Correspondence: D. C. Agourakis (e-mail: dionisio.ca@gmail.com). All 204 theorems in this manuscript and its companion electromagnetic/DSP suite (`MaxwellExterior.lean`) are formally verified from first principles in the Lean 4.15.0 kernel with **zero axioms** (`0 propext, 0 Quot.sound, 0 Classical.choice, 0 sorry`) and deployed in bare-metal `no_std` Rust (`FerriteOS`) and C (`sedenion_pg32_embedded.h`) on an `STM32WB55RG` ARM Cortex-M4F transceiver.

$(w_1 \cdot w_2 = 0)$ be cascaded to null multi-mode interference while allowing a desired signal $s$ to pass through unattenuated? In any associative algebra, $(s \cdot w_1) \cdot w_2 = s \cdot (w_1 \cdot w_2) = 0$ forces total signal collapse. In $\mathbb{S}$, we prove that $(s \cdot w_1) \cdot w_2 = [s, w_1, w_2]$, allowing the desired signal to propagate *purely through the non-associative associator* while two non-coaxial jammers are simultaneously driven to exact zero.

2) **The Cubic Alternativity-Defect Calculus $N_u$:** Why does $u^*(uy) \neq \|u\|_2^2 y$ in $\mathbb{S}$, and can $L(u)$ be inverted or equalized without $\mathcal{O}(16^3)$ matrix inversion or SVD? We prove that left and right alternativity fail by the *exact same symmetric linear operator* $N_u = N_u^\top$, which satisfies the universal degree-3 polynomial $N_u^3 = 4\Lambda(u)^2 N_u$. Every $16\times16$ inverse, MMSE equalizer, Moore–Penrose pseudoinverse $L(u)^+$, and nullspace projector $P_{\ker L(u)}$ reduces to a 3-term closed-form polynomial in $N_u$.
3) **The Co-Axial Zero-Divisor Plane Paradox:** Given two zero divisors $u_1 = e_1 + e_{10}$ and $u_2 = e_3 + e_{12}$, *every* non-zero linear combination $\alpha u_1 + \beta u_2$ in their 2D span is a zero divisor with a 4D nullspace—yet $\ker L(u_1) \cap \ker L(u_2) = \{0\}$! We prove a complete classification of multi-jammer kernel intersections via the $S^6$ box-kite axis fibration $n(u) = (a' \times_{\mathbb{O}} b')/\Lambda(u)$.
4) **Pontryagin's Maximum Principle on $\mathbb{S}$:** How do we maximize the bilinear control Hamiltonian $\mathcal{H}(x, p, J, w) = \langle p, x \cdot w\rangle - \frac{\gamma}{2}\|J \cdot w\|_2^2 - \frac{\rho}{2}\|w\|_2^2$ with respect to the control $w \in \mathbb{S}$ under non-associative jammer dynamics? We prove that the Hessian $Q_{\gamma,\rho}(J) = (\rho + \gamma\|J\|_2^2)I_{16} - \gamma N_J$ is strictly positive definite for all $\rho > 0, \gamma \geq 0$, yielding an exact 3-term SVD-free global maximizer $w^*$.
5) **Eisenstein $S_3$ Phase Symmetry, Sextic Invariants, and Hurwitz–Radon Ternary Codes:** Why is $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$, how can one distinguish lower-octonion $(e_1)$ from twisted-sector $(e_9)$ states that possess identical singular spectra, and how does the 9D Hurwitz–Radon subspace $\mathrm{span}\{e_0, \ldots, e_8\}$ generate both rate-$9/16$ zero-divisor-immune spread spectrum and an extremal $[16, 8, 6]_3$ self-dual ternary code?

## II. The $\mathbb{C} \oplus \mathbb{C}^7$ Model, Convention Census, and $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$

Write $x = (a, b) \in \mathbb{O} \oplus \mathbb{O}e_8$ with imaginary parts $a' = \mathrm{Im}(a) \in \mathbb{R}^7$ and $b' = \mathrm{Im}(b) \in \mathbb{R}^7$. Under the Dickson–Moreno doubling rule

$$(a, b) \cdot (c, d) = \big(ac - d^* b,\ da + bc^*\big), \tag{1}$$

identifying $x \leftrightarrow (\alpha, Z) \in \mathbb{C} \oplus \mathbb{C}^7$ via the complex nucleus coordinate $\alpha = x_0 + ix_8 \in \mathbb{C}$ and the complexified Fano vector $Z = a' + ib' \in \mathbb{C}^7$ (where $i$ acts as right multiplication by $e_8$) collapses 16D sedenion multiplication into a transparent vector identity:

$$(\alpha, Z) \cdot (\beta, W) = \Big(\alpha\beta - \bar{Z} \cdot W,\ \ \bar{\alpha}W + \beta Z + \overline{Z \times_{\mathbb{C}} W}\Big), \tag{2}$$

where $\bar{Z} \cdot W = \sum_{k=1}^7 \bar{Z}_k W_k$ is the Hermitian inner product and $Z \times_{\mathbb{C}} W$ is the $\mathbb{C}$-bilinear extension of the 7D octonionic Fano cross product.

### A. The Quaternion Trap and Opposite-Algebra Isomorphism

In software and hardware implementations of Cayley–Dickson algebras, a recurring source of silent bugs is the choice of conjugation placement in the doubling template $(a, b)(c, d) = (ac - X(b, d),\ Y(a, d) + Z(b, c))$. Across all 64 unital sign-free templates where $X, Y, Z$ independently apply or omit conjugation in either factor order:

- **The Quaternion Trap:** Exactly 16 of the 64 templates produce the valid associative quaternion algebra $\mathbb{H}$ at $\mathbb{C} \to \mathbb{H}$ (because $\mathbb{C}$ is commutative, hiding reversed factor orders), but 14 of those 16 fail norm composition at $\mathbb{H} \to \mathbb{O}$.
- **Uniqueness at $\mathbb{O}$ and $\mathbb{S}$:** Only 2 of the 64 templates yield a normed division algebra at $\mathbb{H} \to \mathbb{O}$: the Dickson–Moreno convention (1) and the Baez–Schafer convention $(ac - db^*, a^* d + cb)$.

**Theorem 1** (Opposite-Algebra Isomorphism and $S_3$ Reflection Automorphism; Lean 4 Theorem 5.44)**.** *Across all* $16 \times 16 = 256$ *basis pairs* $(e_i, e_j) \in \mathbb{S} \times \mathbb{S}$*:*

1) *Conjugation $x \mapsto x^*$ is an exact anti-automorphism $(x \cdot y)^* = y^* \cdot x^*$, establishing that the Baez–Schafer multiplication $x \cdot_{\mathrm{BS}} y = y \cdot_{\mathrm{DM}} x$ is the exact opposite algebra $\mathbb{S}^{\mathrm{op}}$ on the same basis and preserves the $8 + 7$ $\mathrm{PG}(3, 2)$ hyperplane partition identically.*
2) *The reflection $\varepsilon : (a, b) \mapsto (a, -b)$ is an exact algebra automorphism $\varepsilon(x \cdot y) = \varepsilon(x) \cdot \varepsilon(y)$ fixing $\mathbb{O}$ and negating $\mathbb{O}e_8$.*

### B. Why $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$: The Eisenstein Cube-Root Phase Law

By Brown and Schafer's theorem, the automorphism group of the sedenions is $\mathrm{Aut}(\mathbb{S}) \cong G_2 \times S_3$ ($\dim G_2 = 14$). A widespread misconception is that the $S_3$ factor originates from $\mathrm{Spin}(8)$ triality. In fact, octonionic triality is *not* an automorphism of $\mathbb{S}$; the $S_3$ factor comes directly from the $\mathbb{C} \oplus \mathbb{C}^7$ multiplication law (2).

Specifically, consider scaling the complexified Fano vector by a unit phase $Z \mapsto \omega Z$ ($|\omega| = 1$) while fixing $\alpha \in \mathbb{C}$:

- In the scalar slot of (2), $\overline{(\omega Z)} \cdot (\omega W) = |\omega|^2 \bar{Z} \cdot W = \bar{Z} \cdot W$ is invariant for *all* $\omega \in U(1)$.
- In the vector slot, $\bar{\alpha}(\omega W) + \beta(\omega Z) = \omega(\bar{\alpha}W + \beta Z)$, whereas the conjugate cross product transforms as

$$\overline{(\omega Z) \times_{\mathbb{C}} (\omega W)} = \bar{\omega}^2 \overline{Z \times_{\mathbb{C}} W}. \tag{3}$$

Hence $Z \mapsto \omega Z$ is an algebra automorphism of $\mathbb{S}$ if and only if $\bar{\omega}^2 = \omega \iff \omega^3 = 1$!

Setting $\omega = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ yields the explicit order-3 automorphism $\psi \in \mathrm{Aut}(\mathbb{S})$ ($\psi^3 = \mathrm{id}$, $\varepsilon\psi\varepsilon = \psi^{-1}$):

$$\begin{aligned} &\psi(e_0) = e_0, \qquad \psi(e_8) = e_8, \\ &\psi(e_k) = -\frac{1}{2}e_k + \frac{\sqrt{3}}{2}e_{k+8}, \\ &\psi(e_{k+8}) = -\frac{\sqrt{3}}{2}e_k - \frac{1}{2}e_{k+8} \qquad (k = 1, \ldots, 7). \end{aligned} \tag{4}$$

Together with $\varepsilon(Z) = \bar{Z}$, $\langle \varepsilon, \psi\rangle \cong S_3$ commutes elementwise with the diagonal $G_2$ action $(a, b) \mapsto (g(a), g(b))$.

### C. The Sextic $G_2 \times S_3$ Invariant $J_3(u)$

Under $Z \mapsto e^{2\pi i/3}Z$ and $Z \mapsto \bar{Z}$, the complex scalar

$$q(u) \triangleq \bar{Z} \cdot \bar{Z} = \big(\|a'\|_2^2 - \|b'\|_2^2\big) - 2i\langle a', b'\rangle \tag{5}$$

transforms as $q \mapsto e^{-4\pi i/3}q = e^{2\pi i/3}q$ and $q \mapsto \bar{q}$. Consequently, $\mathrm{Re}(q)$ is *not* $S_3$-invariant, and the singular values of $L(u)$ depend only on $|q| = \sqrt{\|a'\|_2^4 + \|b'\|_2^4 + 2\langle a', b'\rangle^2 - 2\Lambda(u)^2}$, making the singular spectrum blind to the sign of $\|a'\|_2^2 - \|b'\|_2^2$. However, cubing $q$ kills the cube root of unity ($e^{6\pi i/3} = 1$), yielding a normalized *sextic $G_2 \times S_3$ invariant*:

$$J_3(u) \triangleq \frac{\mathrm{Re}\big(q(u)^3\big)}{|q(u)|^3} = \frac{\Delta\big(\Delta^2 - 12\langle a', b'\rangle^2\big)}{\big(\Delta^2 + 4\langle a', b'\rangle^2\big)^{3/2}} \in [-1, +1], \tag{6}$$

where $\Delta = \|a'\|_2^2 - \|b'\|_2^2$. As proven in Lean 4 Theorem 5.45, $J_3(u)$ is strictly invariant under all of $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ while cleanly separating lower-octonion basis states ($J_3(e_k) = +1$) from twisted-sector states ($J_3(e_{k+8}) = -1$) that possess identical singular values $\{1^{(16)}\}$.

## III. The Symmetric Alternativity-Defect Calculus $N_u$ and 3-Term SVD-Free Inverses

For any $u = (a, b) \in \mathbb{S}$, define the non-associative wedge magnitude

$$\begin{aligned}\Lambda(u) &\triangleq \|a' \times_{\mathbb{O}} b'\|_2 = \sqrt{\|a'\|_2^2\|b'\|_2^2 - \langle a', b'\rangle^2} \\ &= \tfrac{1}{2}\|\mathcal{A}_{\mathrm{native}}(u)\|_2.\end{aligned} \tag{7}$$

Crucially, *neither $u_0$ nor $u_8$ enters* $\Lambda(u)$: both belong to the associative complex nucleus $\mathbb{C} = \mathrm{span}\{e_0, e_8\}$.

**Theorem 2** (Symmetric Alternativity-Defect Operator $N_u$ and Cubic Minimal Polynomial; Lean 4 Theorems 5.37–5.38)**.** *For every $u = (a, b) \in \mathbb{S}$ and $y = (c, d) \in \mathbb{S}$:*

1) ***Coincidence and Symmetry of Left/Right Alternativity Defects:*** *Left and right alternativity fail by the exact same symmetric linear operator $N_u = N_u^\top : \mathbb{R}^{16} \to \mathbb{R}^{16}$:*

$$\begin{aligned}N_u(y) &\triangleq u(uy) - u^2 y = (yu)u - yu^2 \\ &= \Big([a, b, d]_{\mathbb{O}},\ -[a, b, c]_{\mathbb{O}}\Big),\end{aligned} \tag{8}$$

*where $[a, b, x]_{\mathbb{O}} = (ab)x - a(bx)$ is the octonion associator.*

2) ***Exact Norm-Composition Defect:*** *For all $u, y \in \mathbb{S}$,*

$$\begin{aligned}\|u \cdot y\|_2^2 &= \|u\|_2^2\|y\|_2^2 - \langle y, N_u(y)\rangle \\ &= \|u\|_2^2\|y\|_2^2 + 2\big\langle [a, b, c]_{\mathbb{O}},\ d\big\rangle.\end{aligned} \tag{9}$$

3) ***Normal Equations and Universal Singular Spectrum:*** *Setting $s = \|u\|_2^2$, the Gram operators of left and right multiplication coincide:*

$$L(u)^\top L(u) = R(u)^\top R(u) = sI_{16} - N_u. \tag{10}$$

*Moreover, $N_u$ vanishes on the 8D quaternion-hinge subspace $H_u \oplus H_u e_8$ ($H_u = \mathrm{span}\{e_0, a', b', a' \times_{\mathbb{O}} b'\}$) and satisfies the cubic identity*

$$N_u^3 = 4\Lambda(u)^2 N_u. \tag{11}$$

*Consequently, the eigenvalues of $N_u$ are $\big\{0^{(8)}, +2\Lambda(u)^{(4)}, -2\Lambda(u)^{(4)}\big\}$, and the squared singular values of $L(u)$ are universally*

$$\begin{aligned}\sigma^2\big(L(u)\big) = \Big\{ &s - 2\Lambda(u)\ (\times 4),\ \ s\ (\times 8), \\ &s + 2\Lambda(u)\ (\times 4)\Big\}.\end{aligned} \tag{12}$$

Because $N_u$ satisfies the degree-3 minimal polynomial $X(X^2 - 4\Lambda^2) = 0$, any rational function $f(N_u)$ lies in $\mathrm{span}\{I_{16}, N_u, N_u^2\}$. Table I summarizes the resulting 3-term SVD-free operator formulas. In particular, when $u$ is a Moreno zero divisor ($s = 2\Lambda$), Lagrange interpolation on the non-zero eigenvalues $\{\lambda_1 = s, \lambda_2 = 2s\}$ of $L(u)^\top L(u) = sI_{16} - N_u$ yields the exact integer-scaled Moore–Penrose pseudoinverse

$$32L(u)^+ = \big(8I_{16} - N_u - \tfrac{3}{2}N_u^2\big)L(u^*) \qquad (s = 2,\ \Lambda = 1), \tag{13}$$

and the exact orthogonal projector onto the 4D zero-divisor kernel $\ker L(u) = \ker(N_u - sI_{16})$:

$$P_{\ker L(u)} = \frac{N_u^2 + sN_u}{2s^2}. \tag{14}$$

Restricting Theorem 2 to the unit Moreno cone ($s = 2$, $u^2 = -2e_0$) directly recovers the cone-restricted left-square decomposition $u(u(c, d)) = -2(c, d) + 2(T_1 d, T_2 c)$ of our companion rank–nullity development (`SedenionRank.square_identity` and `SedenionDim.finrank_range_Lu = 12` in [6]), identifying $2(T_1 d, T_2 c) \equiv N_u(c, d)$ while extending the operator calculus unconditionally off the cone to all $u \in \mathbb{S}$.

## IV. Box-Kite Kernel Fibration and the Mutually-Annihilating Cascade Paradox

Every zero divisor $u = (a', b') \in V_2(\mathbb{R}^7)$ defines a unit *box-kite axis* on the 6-sphere $S^6 \subset \mathrm{Im}(\mathbb{O})$:

$$n(u) \triangleq \frac{a' \times_{\mathbb{O}} b'}{\Lambda(u)} = -\frac{\mathcal{A}_{\mathrm{native}}(u)}{\|\mathcal{A}_{\mathrm{native}}(u)\|_2} \in S^6. \tag{15}$$

For any unit frame $(a', b') \in V_2(\mathbb{R}^7)$ and any $p \in \mathrm{Im}(\mathbb{O})$ with $p \perp \{a', b', n(u)\}$, octonion alternativity and anticommutativity yield $(a' \cdot p) \cdot b' = -n(u) \cdot p = p \cdot n(u)$, uniting the Moreno partner map $p \mapsto (a'p)b'$ (`SedenionReal.nullspace` in [6]) with the axis-only parameterization

$$\begin{aligned}\ker L(u) &= \ker R(u) \\ &= \Big\{ (p, -n(u) \cdot p) \;\Big|\; p \in \mathbb{R}^7,\ p \perp \{a', b', n(u)\} \Big\}.\end{aligned} \tag{16}$$

As certified in Lean 4 Theorem 5.45, the 84 primitive zero-divisor blades $e_a \pm e_b$ ($a \in \{1..7\}, b \in \{9..15\}, b \neq a \oplus 8$) fiber over the 14 signed Fano axes $\pm e_m$ ($m = a \oplus b \oplus 8 \in \{1..7\}$) into **14 box-kite sextets** of 6 blades each ($14 \times 6 = 84$).

### A. The Co-Axial Multi-Jammer Nulling Theorem

Suppose a receiver faces two simultaneous zero-divisor jammers $u_1, u_2 \in V_2(\mathbb{R}^7)$. When can a *single* non-zero weight vector $w \in \mathbb{S}$ annihilate both ($u_1 \cdot w = u_2 \cdot w = 0$)?

TABLE I
CLOSED-FORM 3-TERM SVD-FREE OPERATOR CALCULUS IN $\mathbb{S}$ VIA THE SYMMETRIC ALTERNATIVITY-DEFECT OPERATOR $N_u$ ($s = \|u\|_2^2$, $\tau = s + \lambda$, $\Lambda = \|a' \times_{\mathbb{O}} b'\|_2$, $z_0 = u^* \cdot y$)

| **Operator / Physical Task** | **Regime** | **Exact 3-Term Closed-Form Polynomial in $N_u$** | **C-ABI Residual** |
|---|---|---|---|
| Exact Channel Inverse $L(u)^{-1}y$ | $\sigma_{\min}^2 = s - 2\Lambda > 0$ | $\frac{1}{s}z_0 + \frac{1}{s^2-4\Lambda^2}N_u(z_0) + \frac{1}{s(s^2-4\Lambda^2)}N_u^2(z_0)$ | $2.87 \times 10^{-14}$ |
| Tikhonov / MMSE Equalizer $(L^\top L + \lambda I)^{-1}L^\top y$ | $\forall u \in \mathbb{S},\ \lambda > 0$ | $\frac{1}{\tau}z_0 + \frac{1}{\tau^2-4\Lambda^2}N_u(z_0) + \frac{1}{\tau(\tau^2-4\Lambda^2)}N_u^2(z_0)$ | $1.04 \times 10^{-14}$ |
| Moore–Penrose Pseudoinverse $L(u)^+ y$ | $u \in \mathrm{ZD}$ $(s = 2\Lambda)$ | $\frac{1}{s}z_0 - \frac{1}{4s^2}N_u(z_0) - \frac{3}{4s^3}N_u^2(z_0)$ | $1.45 \times 10^{-15}$ |
| Orthogonal Nullspace Projector $P_{\ker L(u)}(x)$ | $u \in \mathrm{ZD}$ $(s = 2\Lambda)$ | $\frac{1}{2s^2}\big(N_u^2(x) + sN_u(x)\big)$ $(\mathrm{rank}\, P_{\ker} = 4,\ L(u)P_{\ker} = 0)$ | $0.00 \times 10^{0}$ |
| Pontryagin Optimal Control $w^* = \arg\max_w \mathcal{H}$ | $\forall J \in \mathbb{S},\ \rho > 0,\ \gamma \geq 0$ | $\frac{1}{\tau_\gamma}\psi + \frac{\gamma}{\tau_\gamma^2-4\gamma^2\Lambda_J^2}N_J(\psi) + \frac{\gamma^2}{\tau_\gamma(\tau_\gamma^2-4\gamma^2\Lambda_J^2)}N_J^2(\psi)$ $(\psi = x^* \cdot p)$ | $1.11 \times 10^{-16}$ |

**Theorem 3** (Co-Axial Kernel Intersection Dichotomy; Lean 4 Theorem 5.39)**.** *For any two zero divisors $u_1, u_2 \in V_2(\mathbb{R}^7)$, the intersection $\ker L(u_1) \cap \ker L(u_2)$ is a complex subspace under right multiplication by $e_8$, so its real dimension belongs strictly to $\{0, 2, 4\}$. Moreover:*

$$\ker L(u_1) \cap \ker L(u_2) \neq \{0\} \iff n(u_1) = +n(u_2). \quad (17)$$

*When $n(u_1) = n(u_2) = n$ and $\mathrm{span}\{a_1', b_1'\} \neq \mathrm{span}\{a_2', b_2'\}$, the intersection is 2-dimensional and spanned by the closed-form joint annihilator*

$$p_0 = (a_1' \times_{\mathbb{O}} a_2') - \langle a_1' \times_{\mathbb{O}} a_2',\, n\rangle n, \qquad w_{\mathrm{joint}} = \big(p_0,\ -n \cdot p_0\big). \quad (18)$$

*Remark* 4 (The Disjoint-Kernel Zero-Divisor Plane Paradox). Consider $u_1 = e_1 + e_{10}$ (axis $n(u_1) = e_1 \times_{\mathbb{O}} e_2 = +e_3$) and $u_2 = e_3 + e_{12}$ (axis $n(u_2) = e_3 \times_{\mathbb{O}} e_4 = +e_7$). For *every* $(\alpha, \beta) \neq (0, 0)$, the linear combination $u(\alpha, \beta) = \alpha u_1 + \beta u_2$ satisfies $\|\alpha e_1 + \beta e_3\|_2 = \|\alpha e_2 + \beta e_4\|_2$ and $\langle \alpha e_1 + \beta e_3, \alpha e_2 + \beta e_4\rangle = 0$; hence *every vector in the 2D plane* $\mathrm{span}\{u_1, u_2\}$ *is a zero divisor with a 4D kernel*. Yet because $n(u_1) = e_3 \neq e_7 = n(u_2)$, Theorem 3 forces $\ker L(u_1) \cap \ker L(u_2) = \{0\}$! No single-stage weight can null both $u_1$ and $u_2$.

### *B. Resolution via Mutually-Annihilating Non-Associative Cascades ($w_1 \cdot w_2 = 0$)*

To defeat two non-coaxial jammers $u_1, u_2$ when $\ker R(u_1) \cap \ker R(u_2) = \{0\}$, we exploit sequential parenthesization:

$$y_{\mathrm{out}} = \big(R \cdot w_1\big) \cdot w_2. \quad (19)$$

Choose $w_1 \in \ker R(u_1)$ $(u_1 \cdot w_1 = 0)$. Because $u_2$ is not co-axial with $u_1$, the Stage-1 residual $u_2' \triangleq u_2 \cdot w_1$ is non-zero. However, when $u_2'$ lies on the zero-divisor cone, $\dim\ker R(u_2') = 4$. Since $\dim\ker L(w_1) = 4$ as well, we can select a Stage-2 weight $w_2 \in \ker R(u_2') \cap \ker L(w_1)$ via the iterated projector $P_{\ker L(w_1)}\big(P_{\ker L(u_2')}(e_k)\big)$!

**Theorem 5** (Mutually-Annihilating 2-Stage Non-Associative Cascade; Lean 4 Theorem 5.39)**.** *Let $u_1 = e_1 + e_{10}$ and $u_2 = e_1 + e_{13}$ be non-coaxial zero-divisor jammers ($n(u_1) = +e_3 \neq -e_4 = n(u_2)$). Define the primitive Stage-1 blade $w_1 = e_4 - e_{15} \in \ker R(u_1)$ and the 4-tap $\{\pm 1\}$ compound Stage-2 zero-divisor blade $w_2 = (e_1 + e_{10}) - (e_6 + e_{13}) \in \ker R(u_2 \cdot w_1) \cap \ker L(w_1)$ (whereas for antipodal axes $u_2 = e_1 - e_{10}$, the 2-tap blade $w_2 = e_1 + e_{10}$ suffices). Then:*

1) ***Mutual Filter Annihilation:*** *The two filter stages annihilate each other:*

$$w_1 \cdot w_2 = 0. \quad (20)$$

2) ***Total Dual-Jammer Cancellation:*** *For any linear combination $J = \alpha u_1 + \beta u_2$,*

$$\big(J \cdot w_1\big) \cdot w_2 = \alpha\big(\underbrace{u_1 \cdot w_1}_{0}\big) \cdot w_2 + \beta\,\underbrace{\big(u_2 \cdot w_1\big) \cdot w_2}_{0} = 0. \quad (21)$$

3) ***Purely Associator-Borne Signal Transmission:*** *For the desired signal $s = e_2$, the cascaded output equals the sedenion associator $[s, w_1, w_2]$ and is strictly non-zero:*

$$\begin{aligned}\big(e_2 \cdot w_1\big) \cdot w_2 &= e_2 \cdot \underbrace{\big(w_1 \cdot w_2\big)}_{0} + \big[e_2, w_1, w_2\big] \\ &= \big[e_2, w_1, w_2\big], \quad \big\|\big(e_2 \cdot w_1\big) \cdot w_2\big\|_2^2 = 4.\end{aligned} \quad (22)$$

Theorem 5 exhibits a phenomenon with no counterpart in linear or associative algebra: the effective end-to-end transfer operator $T(x) = (x \cdot w_1) \cdot w_2 = [x, w_1, w_2]$ is *100% non-associative*, vanishing identically if the underlying algebra were associative, yet delivering full rank-8 signal transmission while zeroing two non-coaxial jammers.

## V. PONTRYAGIN'S MAXIMUM PRINCIPLE OVER $\mathbb{S}$ AND $\mathrm{PG}(3, 2)$

Consider a controlled 16D sedenion RF state trajectory $\dot{x} = f(x) + x \cdot w$ with state $x \in \mathbb{S}$, costate (adjoint vector) $p \in \mathbb{S}$, jammer state $J \in \mathbb{S}$ (weighted by $\gamma \geq 0$), and control effort weight $\rho > 0$. By Pontryagin's Maximum Principle (PMP) [3], the optimal control $w^* \in \mathcal{U} \subseteq \mathbb{S}$ maximizes the *Pontryagin Control Hamiltonian*:

$$\mathcal{H}(x, p, J, w) \triangleq \langle p, x \cdot w\rangle - \frac{\gamma}{2}\|J \cdot w\|_2^2 - \frac{\rho}{2}\|w\|_2^2. \quad (23)$$

**Theorem 6** (Closed-Form Sedenion Pontryagin Maximum Principle; Lean 4 Theorems 5.42–5.43)**.** *For any state $x \in \mathbb{S}$, costate $p \in \mathbb{S}$, jammer $J \in \mathbb{S}$, jammer penalty $\gamma \geq 0$, and effort weight $\rho > 0$:*

1) ***Adjoint Switching Reduction and Strict Concavity:*** *By the Cayley–Dickson adjoint identity $L(x)^\top = L(x^*)$ and Theorem 2, define the* Pontryagin switching vector *$\psi(x, p) \triangleq x^* \cdot p$ and the symmetric Hessian operator*

$$Q_{\gamma,\rho}(J) \triangleq \big(\rho + \gamma\|J\|_2^2\big)I_{16} - \gamma N_J. \quad (24)$$

*Then $\mathcal{H}(x, p, J, w) = \langle \psi, w\rangle - \frac{1}{2}\langle w, Q_{\gamma,\rho}(J)w\rangle$. Since $2\Lambda(J) \leq \|J\|_2^2$, the minimum eigenvalue of $Q_{\gamma,\rho}(J)$ is $\rho + \gamma\big(\|J\|_2^2 - 2\Lambda(J)\big) \geq \rho > 0$; thus $\mathcal{H}$ is strictly concave on $\mathbb{R}^{16}$.*

2) ***3-Term SVD-Free Unconstrained Global Maximizer*** *($\mathcal{U} = \mathbb{S}$): Let $\tau = \rho + \gamma\|J\|_2^2$. The unique global maximizer $w^* = \arg\max_{w\in\mathbb{S}} \mathcal{H}(x,p,J,w) = Q_{\gamma,\rho}(J)^{-1}\psi$ is given in closed 3-term SVD-free form by*

$$w^* = \frac{1}{\tau}\psi + \frac{\gamma}{\tau^2 - 4\gamma^2\Lambda(J)^2} N_J(\psi) + \frac{\gamma^2}{\tau\big(\tau^2 - 4\gamma^2\Lambda(J)^2\big)} N_J^2(\psi), \tag{25}$$

*achieving the global maximum $\mathcal{H}(x,p,J,w^*) = \frac{1}{2}\langle\psi, w^*\rangle$, with exact perturbation deficit $\mathcal{H}(w^*) - \mathcal{H}(w^* + \delta) = \frac{1}{2}\langle\delta, Q_{\gamma,\rho}(J)\delta\rangle > 0$ for all $\delta \neq 0$.*

3) ***Zero-Divisor Unit-Sphere Maximizer*** *($\gamma \to \infty$, $\mathcal{U} = \{w \in \ker L(J) : \|w\|_2 = 1\}$): When $J \in V_2(\mathbb{R}^7)$ is a Moreno zero divisor ($\|J\|_2^2 = 2\Lambda(J)$), the constrained unit-norm control maximizing $\langle p, x\cdot w\rangle$ subject to exact jammer annihilation $J \cdot w = 0$ is*

$$w^*_{\ker} = \frac{P_{\ker L(J)}(x^*\cdot p)}{\big\|P_{\ker L(J)}(x^*\cdot p)\big\|_2}, \qquad \mathcal{H}^*_{\ker} = \big\|P_{\ker L(J)}(x^*\cdot p)\big\|_2. \tag{26}$$

When the control is restricted to the discrete 168-element Pre-ADC analog crossbar alphabet $\mathcal{B}_{84}^{\pm} = \{\pm\frac{1}{\sqrt{2}}(e_a \pm e_b)\}$, Pontryagin maximization reduces to the finite bang-bang selection $w^*_{\mathrm{BB}} = \arg\max_{w\in\mathcal{B}_{84}^{\pm}} \mathcal{H}(x,p,J,w)$, which automatically selects one of the 4 zero-divisor partners of $J$ whenever $\gamma \gg 1$.

## VI. Hurwitz–Radon 9-Frame Isometries and Extremal $[16,8,6]_3$ Self-Dual Ternary FEC

Although $\mathbb{S}$ contains a 10D sphere of unit zero divisors (and 8 maximal zero-divisor-free octonionic coordinate cells characterized combinatorially via the Sedenion Triangular Root [4], [5]), Hurwitz and Radon's theorem on sums of squares ($\rho(16) = 9$) implies that $\mathbb{S}$ also houses a 9-dimensional linear subspace of *universal norm-preserving multipliers*.

**Theorem 7** (Hurwitz–Radon 9-Frame Isometry and Extremal $[16,8,6]_3$ Self-Dual Ternary Code; Lean 4 Theorems 5.40–5.41)**.** *Define the 9-dimensional Hurwitz–Radon subspace $\mathcal{V}_9 \triangleq \mathrm{span}\{e_0, e_1, \ldots, e_8\} \subset \mathbb{S}$. Then:*

1) ***Universal Norm Composition Against Arbitrary Keys:*** *For every $u, v \in \mathcal{V}_9$ with $u \perp v$ and* every *$x \in \mathbb{S}$ (including singular Moreno zero divisors $x \in V_2(\mathbb{R}^7)$),*

$$L(u)^\top L(v) + L(v)^\top L(u) = 0, \qquad \|u\cdot x\|_2^2 = \|x\cdot u\|_2^2 = \|u\|_2^2\|x\|_2^2. \tag{27}$$

*Consequently, spreading a 9-symbol vector $\boldsymbol{m} \in \mathbb{R}^9$ into $\mathbb{S}$ via $y = \big(\sum_{i=0}^{8} m_i e_i\big)\cdot w$ over any non-zero key $w \in \mathbb{S}$ admits exact inversion-free despreading $m_i = \langle e_i \cdot w, y\rangle / \|w\|_2^2$.*

2) ***Extremal*** *$[16,8,6]_3$* ***Self-Dual Ternary Code:*** *Reducing the Hurwitz–Radon sum $\sum_{i=1}^{8} L(e_i) \bmod 3$ yields a systematic generator matrix $G_3 = [I_8 \mid P_8] \in \mathbb{F}_3^{8\times 16}$ satisfying exact self-duality $G_3 G_3^\top \equiv 0 \pmod 3$ and minimum Hamming distance $d_{\min} = 6$ (achieving the Mallows–Sloane bound $3\lfloor 16/12\rfloor + 3 = 6$). Its 8-trit syndromes uniquely distinguish all $1 + 32 + 480 = 513$ error patterns of Hamming weight $\leq 2$, guaranteeing bounded-distance correction of any 1- or 2-trit error.*

## VII. Zero-Axiom Lean 4 Verification and Standalone Cortex-M4F Hardware Deployment

Every algebraic theorem in this paper has been formalized from first principles in Lean 4.15.0 (`proofs/MaxwellExterior.lean` and `proofs/Axioms.lean`) using a constructive formal-difference integer kernel (`MaxwellExterior.Z`) that eliminates all standard-library axiom dependencies (`204/204` theorems report `does not depend on any axioms` under `#print axioms`). Combined with the companion `sedenion-pg32` formalization (`172/172` zero-axiom theorems in `lean/pg32/` and `140` real-analytic Mathlib theorems in `lean/real/` [6]), the unified suite comprises **516 machine-checked Lean 4.15.0 theorems** (`0 sorry`), of which **376 depend on zero axioms**.

On the hardware side, all five constructions execute natively in bare-metal `no_std` Rust (`FerriteOS/crates/ferrite-rf/src/sedenion_pg32.rs`) and standalone C (`sedenion_pg32_embedded.h`) on an `STM32WB55RG` 64 MHz ARM Cortex-M4F transceiver (deployed as standalone `sedenion_pg32.fap` and `ferrite_field.fap` applications on the Flipper Zero microSD card), achieving:

- **3-Term SVD-Free $N_u$ MMSE & Moore–Penrose Equalization:** 0.0–0.1 ppm single-precision `f32` reconstruction error on Cortex-M4F ($2.87 \times 10^{-14}$ in `f64` C-ABI) with 0 B heap allocation.
- **Co-Axial & Mutually-Annihilating Cascades:** $0.00 \times 10^0$ dual-jammer residual and $w_1 \cdot w_2 = 0$ with $\|(e_2 \cdot w_1)\cdot w_2\|_2 = 2.0$.
- **Pontryagin Maximum Principle Solver:** Real-time closed-form 3-term evaluation of $w^* = \arg\max_w \mathcal{H}(x,p,J,w)$ ($\mathcal{H}^* = 0.8144$ unconstrained; $\mathcal{H}^*_{\ker} = 3.9206$ with 0.0 jammer leakage on the zero-divisor sphere).
- **On-Chip $[16,8,6]_3$ Ternary FEC Sweep:** Exhaustive on-device verification of all $513/513$ coset leaders of weight $\leq 2$ on the 64 MHz Cortex-M4F.

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